



ENGINEERING ANALYSIS

# Recognition and Movement Pipeline Analysis

CPU runtime, GPU research, reacquisition, and native X/Y evidence.

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## Historical bounded speed-sensitive replacement - 7 September 2026

The former bounded native motion experiment used completed MediaPipe palm observations and separates resting noise from intentional velocity. It measures consecutive MediaPipe coordinates using capture timestamps and normalizes velocity by the player's directional reach. Calibration noise creates an elliptical X/Y resting region with hysteresis. Above it, one coherent two-dimensional newest-coordinate weight rises smoothly from [0.70](#) to [1.00](#) at 1.50 calibrated reach spans per second. Large travel and meaningful reversals are therefore direct. A stop settles inside the measured noise region on its first fresh result and exactly on the next, avoiding independent per-axis snaps.

The runtime cap is unconditionally [1.00](#). Historical [motion\\_coordinate\\_boost](#) and [motion\\_coordinate\\_max](#) fields remain loadable, but cannot create extrapolation. No moving-average window, queue, replay, prediction, or core-side filter was added.

MediaPipe is the only live coordinate source because optical-flow movement was reported as jerky and unreliable even though the underlying MediaPipe tracking remained usable. The optical-flow implementation is retained as inactive source and historical trace support, but [motion\\_tracking](#) no longer enables it. The Dashboard no longer exposes the bounded curve; live movement uses direct Latest coordinates.

The initial MediaPipe-first deployment exposed two calibration mistakes in the bounded curve. Its [0.70](#) slow-follow weight was referenced to 60 Hz even though the Controller measured roughly 9-10 completed inferences per second; temporal normalization therefore raised most real samples to nearly one-to-one. The saved test calibration also contained saturated [1.0](#) jitter values, which the curve interpreted as a large image-space dead zone. Follow weighting now uses a 100 ms reference interval, and saturated jitter falls back to the small fixed floor while preserving the saved center, scale, and wrist.

[benchmark-vision-replay.py](#) version 2 retains timestamps and cue definitions in its aggregate local report. [benchmark-native-motion-curve.py](#) consumes that report, compares the former [.70 / 15 / 1.30](#) experiment, a capped error-driven reference, the actual speed curve, and direct latest coordinates, then evaluates 27 bounded candidates. It reports neutral span, selected-to-filtered error, medium 90% response, large first-sample misses, reversals, overshoot, continuity, and recognition age when present. The temporary physical clip is not currently available on the development Mac, so recorded-clip and synchronized live camera-to-display validation remain pending.

The 0.4.0 production choice is Latest coordinate. It publishes every valid reach-clamped MediaPipe point directly during continuous tracking. After a brief dropout, one result that contradicts established motion-or is unusually distant without strong forward alignment-is held until the next fresh result. Live tracing showed that allowing strongly aligned forward recovery immediately avoided the earlier compound catch-up jump while inserting no unnecessary holds in the tested gameplay window. This is a reacquisition guard, not smoothing or prediction.

The runtime now rejects malformed/non-finite landmark geometry, retains the calibration-compatible five-point wrist/knuckle average as the production anchor, and exports three alternative anchors for comparison. It clamps the selected point to player reach before processing. The historical comparison applied the same clamp to both Latest and Bounded lanes. Runtime clears history on stale/lost input and uses the frame capture timestamp rather than inference-start time for freshness.

The gameplay and dot recordings contain aggregate timings and sampled validity; they do not retain recognized, flow, selected, and filtered coordinates for each individual move. They cannot establish how many camera images or presented game frames contained intermediate hand positions. Receiver publications are not necessarily new measurements or different coordinates.

## Observed evidence

| 30-second window                        | Valid Controller samples | Observed tracking losses |
|-----------------------------------------|--------------------------|--------------------------|
| Earlier flow revision, dot movement     | 579/591 (98.0%)          | 6                        |
| Recognition fallback, fast dot movement | 447/589 (75.9%)          | 34                       |
| Recognition fallback, actual gameplay   | 568/590 (96.3%)          | 3                        |

A later MediaPipe-only, Dashboard-closed status run measured 96.5% detected samples, 65.4 ms median and 76.2 ms p95 source age, 52.2 ms median and 58.3 ms p95 inference, 2 ms p95 send work, and about 16 distinct native updates per second. This is a software-stage sample, not physical hand-to-display latency.

These were different physical movements, not controlled before/after trials. In gameplay, 395/590 polls reported `flow_unavailable`, 61 reported `flow_seed_unavailable`, and one reported `correction_budget`. Fallback therefore frequently delivered recognition coordinates rather than a newer flow estimate. These are sampled frame counts, not distinct recognition-result counts. The receiver observed 1,367 distinct valid publications in 30 seconds, not 1,367 new recognized positions or displayed images.

## Recognition runtime and Adreno 702 research - 9 September 2026

### What the MediaPipe graph means

The MediaPipe graph is the complete hand-processing pipeline, not one neural network. It connects frame preparation, palm detection, hand-region geometry, the landmark model, tracking between detections, and result delivery. The palm and landmark neural networks are nodes inside that graph. During continuous tracking, the landmark model uses the previous hand region. When that evidence fails, the graph runs the more expensive palm detector to reacquire the hand.

PowerGlove Vision consumes the graph's 21 landmarks once per fresh result. The five-point palm anchor and gesture calculations happen after MediaPipe has already calculated all landmarks; reducing the number of points averaged by PowerGlove Vision would therefore not reduce neural-network work.

### MediaPipe 0.10.35 CPU promotion

The release now ships only the ARM64/Python 3.12 MediaPipe 0.10.35 wheel. The production lane remains the complete MediaPipe Hands graph with four XNNPACK CPU threads, fused full-colour preparation, a 0.35 tracking-confidence threshold, the 2.25 hand-search region, and Latest-coordinate native X/Y. Gesture rules, calibration, reach, mappings, and controller transport did not change.

The same retained 736-frame fast-sweep clip was replayed through the former 0.10.18 runtime and the selected 0.10.35 runtime with identical settings:

| Runtime            | Inference p50 / p95 | Detection continuity | One-frame misses |
|--------------------|---------------------|----------------------|------------------|
| Historical 0.10.18 | 36.12 / 60.18 ms    | 97.96%               | 15               |
| Shipped 0.10.35    | 34.03 / 46.60 ms    | 97.96%               | 15               |

Landmark-continuation p95 fell from 51.10 ms to 44.29 ms. Palm-reacquisition p95 fell from 149.48 ms to 120.16 ms. A separate live Super Glove Ball trace improved capture-to-send p95 from 126.84 ms to 91.16 ms and delivered all 3,408 observed samples to RetroPie with no trace drops. The player reported that both ordinary tracking and return-to-space reacquisition felt zippier.

An output-paused ten-minute soak exercised the complete camera and graph. It completed without a crash, camera loss, worker error, or thermal throttling. The hottest reported thermal zone reached 61.7 degrees Celsius. Container memory rose during initialization, then remained at an approximately 484 MiB plateau. The blank-room portion deliberately stressed repeated palm detection; the matched clip supplies the continuity evidence that an unattended room cannot.

The packaged wheel was rebuilt from upstream MediaPipe commit [f8ef212d5c962c0e853db7e59d217056b187084b](#). Its release metadata removes unused JAX and JAXLIB dependencies and pins headless OpenCV 4.11.0.86. The rebuilt integrity record has SHA-256 [3f09815d9f6c41d828cd71c9ba477c24a63850908876dfc8b095a882c790e562](#). An isolated ARM64 import confirmed MediaPipe 0.10.35+powerglove.gpu2, OpenCV 4.11.0, and no installed JAX module.

## Why production still uses the CPU

The UNO Q exposes a genuine Adreno 702 at 845 MHz through Mesa's Turnip driver. Tests reached the hardware through EGL/OpenGL ES, OpenCL through Rusticl, and Vulkan. Hardware access was therefore confirmed; the limitation was the tested inference software path, not an absent GPU.

| Exact or representative lane                 | CPU result     | Adreno result    | Decision                                   |
|----------------------------------------------|----------------|------------------|--------------------------------------------|
| MediaPipe 0.10.35 hand landmark continuation | 32.74 ms p50   | 384.12 ms p50    | GPU rejected                               |
| MNN 3.6.1 landmark-lite model                | 19.7 / 26.8 ms | 43.9 / 44.0 ms   | Vulkan slower and numerically incompatible |
| MNN 3.6.1 palm-lite model                    | 36.3 / 44.3 ms | 77.9 / 78.2 ms   | Vulkan slower and numerically incompatible |
| ncnn exact landmark model                    | 25.0 / 38.7 ms | 385.5 / 386.9 ms | Vulkan slower and numerically incompatible |

The MediaPipe OpenGL graph repeatedly synchronized small tensors and was more than ten times slower than XNNPACK. The experimental Linux OpenCL port reached real FD702 graph construction but failed while creating an image from a buffer. MNN's per-layer profiler showed time distributed across convolution and depthwise-convolution work rather than one fixable synchronization tail.

An ncnn CPU sidecar did reproduce the exact landmark model and closely matched MediaPipe coordinates, but its complete replay measured 36.1-36.5 ms p50 and 45.7-52.3 ms p95 with 98.68% continuity. Image preparation, process handoff, and duplicated tracking geometry removed its model-only advantage. It remains a repository research tool rather than an installed backend.

Qualcomm QNN remains a possible future lane only if a redistributable UNO Q runtime becomes available and proves end-to-end improvement. No confidence, reach, or gesture tuning can repair hundreds of milliseconds spent inside a delegate. Any future accelerator must beat the current camera-to-coordinate p95 by at least 20%, preserve ordered newest-sample delivery and recognition within one percentage point, add no jitter or false gestures, and remain thermally stable for ten minutes.

The supporting upstream references are MediaPipe's framework documentation, hand-tracking graphs and model documentation; Qualcomm's AI Engine Direct and TensorFlow Lite delegate documentation; MNN's Vulkan backend;

ncnn's Vulkan documentation; and Mesa's Freedreno/Turnip device support. Exact commits, converted-model hashes, profiler outputs, and temporary binaries remain engineering evidence and are deliberately excluded from ordinary installers.

## Historical smoothing-only experiment

Before the speed-curve replacement, [scripts/analyze-motion-samples.py](#) exercised the error-driven native movement engine with an ideal instantaneous measured-position step. It assumed 60 updates/second, no input noise or loss, and the former code defaults (minimum .70, maximum 1.00, motion boost 4.00). The following table is retained as historical evidence for that implementation.

| Step in normalized camera X | Time from first step sample to 95% of target |
|-----------------------------|----------------------------------------------|
| 0.010                       | 200 ms                                       |
| 0.025                       | 200 ms                                       |
| 0.050                       | 167 ms                                       |
| 0.100                       | Immediate on first sample                    |
| 0.200                       | Immediate on first sample                    |

With smoothing disabled, every modeled step passes through on its first sample. That comparison does **not** measure the jitter cost of disabling smoothing. The elapsed times exclude capture, recognition, transport, game simulation and display. They must not be added to independent timing percentiles. The final five-percent criterion includes a small settling tail; it is not a delay before movement begins.

The speed-dependent behavior explained the problem: the former coefficient used position error, then adjusted for elapsed time. It was adaptive, but was not a One Euro velocity-based filter. Large errors can bypass damping while small aiming corrections retain it. This is a plausible contributor to the reported difficulty catching and directing the ball, not proof of the complete cause.

The script now exercises the bounded speed curve and labels its parameters in new reports. Prediction accuracy cannot be tested from these aggregate reports. A predictor can extrapolate an old measurement, but abrupt stops and reversals can cause overshoot. Do not fit or enable prediction using these reports as ground truth.

## Prepared trace extension

The optional finite diagnostic trace now includes, on each vision event:

- New recognition completion, its original capture timestamp, X/Y, confidence, and detection flag. In-flight frames contain no fabricated recognition event.
- Accepted anchor's original capture timestamp.
- Attempted optical-flow X/Y and a separate acceptance flag; a correction that exceeded its budget must not be counted as accepted flow.
- Selected normalized X/Y, fallback reason, and invalid-input reason.
- Final filtered normalized X/Y, smoothing parameters, and mapped native axes.

Capture sequence/timestamp and transport sequence remain available for correlation. Raw and filtered normalized coordinates share camera space; native axes use the player's reach mapping. Invalid selected coordinates are null. A repeated source is identifiable by anchor timestamp and must not be treated as a fresh velocity measurement. The existing trace remains opt-in, finite, buffered, and asynchronous. No video is recorded by this extension.

The instrumentation was subsequently deployed with the medium-jump trial below; trace collection has not yet been enabled. Before the next physical trial, enable a finite trace, then record the real hand and cabinet screen together with the iPhone. Test small corrections, a large move, a stop, and a reversal. Trace replay can compare smoothing outputs, while video supplies physical timing and overshoot evidence. Cross-device clocks require alignment; do not subtract them directly.

Reproduce the offline report with:

```
SH
python3 scripts/analyze-motion-samples.py \
  --samples-dir /tmp/uno-dot-baseline-x8heur1f \
  --output /tmp/new-motion-analysis.json
```

Analyze one finite per-frame trace without replaying controller input:

```
SH
python3 scripts/analyze-motion-trace.py /tmp/controller.trace.json \
  --output /tmp/controller-motion-analysis.json
```

For a controlled directory whose filenames follow `min0.70-boost8.trace.json`, compare every configuration with:

```
SH
python3 scripts/compare-motion-matrix.py /tmp/controller-motion-matrix \
  --output /tmp/controller-motion-matrix.json
```

Both tools create new reports rather than overwriting existing evidence. Their settling estimates describe normalized software coordinates, not physical hand-to-screen latency.

## Medium-jump and bounded extrapolation trial

This historical experiment let `motion_coordinate_boost` override the boost in `update_native_motion`; null preserved the baseline value. The Controller trial used boost 15.0 instead of the baseline 4.0, with minimum .70. An additional experimental `motion_coordinate_max` cap of 1.30 allowed bounded extrapolation beyond the newest measured position. This was an error relative to the filtered position, not a speed threshold or a percentage of the game screen. It does not remove recognition age.

The actual-engine 60 Hz step model gives 0 ms additional settling time for .04 and .05 steps with boost 8, compared with 183 and 167 ms respectively at boost 4. A .005 step remains damped (200 ms to 95%, previously 217 ms). Noise and physical response still require the paired recording. Baseline synchronous tracking was unchanged. The bounded speed-sensitive replacement now ignores these legacy overrides during native movement while continuing to load them. Trace collection must be explicitly enabled for a physical test.

## Six-configuration trace matrix

The matrix varied minimum smoothing and experimental motion boost while holding the 250 ms freshness limit, 320 px flow width, 25 ms correction budget, and maximum smoothing 1.00 fixed. Each window produced a finite trace with zero dropped records. The analysis uses [scripts/compare-motion-matrix.py](#).

| Minimum / boost | Vision events | Valid % | Losses | Age p95 (ms) | Medium settling p95 (ms) |
|-----------------|---------------|---------|--------|--------------|--------------------------|
| 0.70 / 4        | 834           | 69.18   | 14     | 104.5        | 0.0                      |
| 0.70 / 8        | 751           | 56.72   | 17     | 175.0        | 0.0                      |
| 0.70 / 12       | 895           | 90.17   | 7      | 87.2         | 84.6                     |
| 0.85 / 4        | 884           | 89.37   | 4      | 104.1        | 114.0                    |
| 0.85 / 8        | 886           | 97.74   | 1      | 85.4         | 96.9                     |
| 0.85 / 12       | 897           | 97.44   | 2      | 83.1         | 63.8                     |

These windows did not contain a controlled, repeated movement sequence, so the rows cannot select a production candidate. Validity and recognition age varied enough to confound settling comparisons. The results are a harness check and preliminary range, not proof that 0.85 / 12 is better. The live setting was restored to 0.70 / 8; calibration was unchanged.

## Controlled three-window follow-up

The follow-up repeated the same center, small-correction, medium-jump, reversal, and center sequence for the baseline and two candidates. All three finite traces had zero dropped records.

| Configuration       | Vision events | Valid % | Losses | Age p95 (ms) | X error median / p95 | Medium settling p95 (ms) |
|---------------------|---------------|---------|--------|--------------|----------------------|--------------------------|
| 0.70 / 8 (baseline) | 778           | 94.47   | 6      | 87.5         | 0.0011 / 0.0122      | 120.8                    |
| 0.85 / 8            | 883           | 92.87   | 5      | 91.5         | 0.0006 / 0.0052      | 17.9                     |
| 0.85 / 12           | 894           | 90.27   | 7      | 111.5        | 0.0004 / 0.0041      | 112.1                    |

The controlled follow-up showed **0.85 / 8** was a useful intermediate smoothing candidate, but later gameplay testing found it too floaty. The live experiment was returned to minimum smoothing 0.70 and then raised to boost 15 with cap 1.30. Those later values are a user-driven exploratory setting, not a production default; repeat controlled traces before promoting them.